Approval request and prerequisites

Measurement of $\nu_\mu + \bar{\nu}_\mu$ charged-current single-pion production on argon in the NuMI beam with the MicroBooNE detector

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This document contains the decision requested from the collaboration, the status of every result with the prerequisites that apply to it, the validation ladder, and the request to compare the control regions with beam-on data. The analysis is described in the analysis note, the proton-tagged note and the technical supplement.

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1 Decision requested

1. Approve the inclusive CC$\pi^\pm$ signal definition, selection, extraction and uncertainty model, and the reported inclusive observables, each as a Primary result: $d\sigma/dp_\mu$, $d\sigma/d\cos\theta_\mu$, $d\sigma/d\cos\theta_\pi$, $d\sigma/d\theta_{\mu\pi}$, the cross section in three p_π regions and the one-bin total cross section, each in the FHC, RHC and combined configurations.
2. Do not yet authorise opening of the inclusive signal region: the prerequisites U1–U6 of Table 4 remain to be completed.
3. Approve the following, each as a Secondary result: the proton-tagged one-bin total cross section and the proton-tagged $d\sigma/d\theta_p$ and $d\sigma/d\theta_{\pi p}$ (FHC, RHC, combined), and the two-dimensional results $d^2\sigma/d\theta_p\, d\delta p_T$ (proton-tagged sample) and $d^2\sigma/d\cos\theta_\pi\, d\cos\theta_\mu$ (inclusive sample) in the combined configuration only, subject to the prerequisites listed for each in Table 4.
4. Treat every other proton-tagged extraction (p_p, W_{had}, δp_T, $\delta\alpha_T$, $\delta\phi_T$, p_n and the muon and pion observables on the proton-tagged sample) as a Validation-only product.
5. Do not report the differential $W_{\pi p}$ result, the other two-dimensional pairs, or the FHC and RHC extractions of the two two-dimensional results (Not reported).

2 Status of the results

Tables 1 and 2 give the status of every inclusive and proton-tagged product and the prerequisites of Table 4 that apply to it.

Table 1: Status of the inclusive products. "yes" marks the configurations in which a product is reported. The prerequisites are those of Table 4.

Product	FHC	RHC	Comb.	Status	Prerequisites
$d\sigma/dp_\mu$, $d\sigma/d\cos\theta_\mu$, $d\sigma/d\cos\theta_\pi$, $d\sigma/d\theta_{\mu\pi}$, σ in three p_π regions	yes	yes	yes	Primary result	U1–U7
One-bin total cross section	yes	yes	yes	Primary result	U1–U3, U5–U7
$d^2\sigma/d\cos\theta_\pi\, d\cos\theta_\mu$	no	no	yes	Secondary result	U1–U3, U5–U7, R1
$d\sigma/dp_\pi$ in five bins				Not reported	
$d\sigma/d\theta_\mu$				Not reported	
Two-dimensional $\cos\theta_\pi \times \theta_{\mu\pi}$, $\cos\theta_\mu \times p_\mu$, $\cos\theta_\pi \times p_\pi$; FHC and RHC $\cos\theta_\pi \times \cos\theta_\mu$				Not reported	

Table 2: Status of the proton-tagged products. "yes" marks the configurations in which a product is reported. The prerequisites are those of Table 4.

Product	FHC	RHC	Comb.	Status	Prerequisites
One-bin total cross section	yes	yes	yes	Secondary result	U1–U3, U5–U7, P1
$d\sigma/d\theta_p$, $d\sigma/d\theta_{\pi p}$	yes	yes	yes	Secondary result	U1–U3, U5–U7, P1, P2
$d^2\sigma/d\theta_p\, d\delta p_T$	no	no	yes	Secondary result	U1–U7, P1–P3, R1
$d\sigma/dp_p$	yes	yes	yes	Validation-only product	P1
$d\sigma/dW_{\text{had}}$ (two regions)	yes	yes	yes	Validation-only product	P1, P4
$d\sigma/d\delta p_T$, $d\sigma/dp_n$	yes	yes	yes	Validation-only product	P1, P3, P4
$d\sigma/d\delta\alpha_T$, $d\sigma/d\delta\phi_T$	yes	yes	yes	Validation-only product	P1
p_μ, p_π (two regions), $\cos\theta_\mu$, $\cos\theta_\pi$, $\theta_{\mu\pi}$ on the proton-tagged sample	yes	yes	yes	Validation-only product	P1
$d\sigma/dW_{\pi p}$				Not reported	
Two-dimensional $\theta_p \times W_{\pi p}$, $\theta_{\pi p} \times W_{\pi p}$, $\theta_{\pi p} \times \delta p_T$; FHC and RHC $\theta_p \times \delta p_T$				Not reported	

3 Validation ladder

Table 3 lists the validation steps with their status and criteria. Closure, the ensembles and the reweighted-model test are complete for the primary and the secondary results; the independent-generator closure (U3), the control-region comparison (U2, and P1 for the proton mis-assignment) and the prerequisites for opening the signal region are not.

Table 3: Validation steps of the extraction. A result can close under the nominal model while it is determined largely by the modelled migrations or by the regularisation; the migration criterion and the trace of A_C test that separately (Sec. 11.3 of the analysis note).

Step	Test	Status	Criterion	Largest deviation	Required before
1. Implementation closure	Does the chain return the truth of pseudo-data thrown from the model of the response?	Done for every extraction	Unfolded result consistent with the A_C-smeared truth; integral ratio within the statistical spread	Inclusive: $\chi^2 \leq 3.12/8$, integral ratios 0.905–1.106. Proton-tagged: $\chi^2 \leq 7.28/4$ (RHC $\cos\theta_\pi$), ratios 0.91–1.21; two-dimensional: cell ratios 0.38–1.48	Unblinding

Step	Test	Status	Criterion	Largest deviation	Required before
2. Statistical calibration	Are the quoted statistical uncertainties correctly sized?	Done: seven inclusive and ten proton-tagged extractions, 100 pseudo-experiments each, among them θ_p and $\theta_{\pi p}$ in every configuration and the two two-dimensional results (V1)	Pull width consistent with one within the ensemble precision ($\pm 13\%$ at 100 members); coverage 68% and 95%	Inclusive: widths 0.92–1.04, 68% coverage 64–72%; the sparsest two-dimensional cell lies 0.57 statistical standard deviations low on average. Proton-tagged: widths 0.89–1.04, 66–73%	Unblinding
3. Parametric robustness	Does the extraction follow a different model that reweighting can reach?	Done: GENIE multisim universe 545 and $\Delta \to N\pi$ angular, inclusive FHC and RHC, proton-tagged FHC including θ_p and $\theta_{\pi p}$, and the two-dimensional results in the combined configuration (V1)	Per-bin bias against the per-bin total uncertainty; a bias above 1σ is flagged and treated	Inclusive: 1.21σ in one sparse bin (FHC $\cos\theta_\mu$). Proton-tagged: -1.19σ and -1.02σ in the low δp_T and p_n bins (P3), $+1.04\sigma$ in the lower W_{had} region. θ_p, $\theta_{\pi p}$ and the two-dimensional results: at most 0.91σ	Unblinding
4. Structural robustness	Does it close on pseudo-data from a generator that reweighting cannot reach?	Outstanding (U3)	Criterion of U3	n/a	Unblinding
5. Background validation	Do control-region data support the background model?	Outstanding (U2; P1 for the proton mis-assignment)	Frozen procedure of the control regions	n/a	Unblinding
6. Signal-region opening	Are the normalisation and covariance prerequisites complete?	Outstanding	U1–U6 complete	n/a	n/a

4 Prerequisites

Inclusive results. The inclusive signal region can be opened once U1–U6 are complete; U7 is required before publication, and U4 applies to the results that use the muon momentum. The flux confirmation (U1) concerns the three integrated fluxes of Table 6 of the analysis note: the NuMI flux group is asked to confirm in writing that they are the PPFX-corrected central values that correspond to the per-event PPFX weights of the simulation (Sec. 4.4 of the analysis note). The beam-on exposure check (U5) replaces the recorded exposures of Table 5 of the analysis note by values summed over the runs contained in each beam-on file. The MCS and range momenta of contained muons agree between data and simulation within 1.5% up to 3 GeV/c; the MCS momentum of exiting muons, lower in the data, is calibrated to the control regions from version 1.9 (U4, complete; Sec. 12.3 of the analysis note). The clean-cache reproduction of Sec. 22 of the technical supplement is to be repeated on the release to be unblinded (U6), and the three terms outside the official covariance (Sec. 12.4 of the analysis note) are to be included before publication (U7).

Independent-generator closure (U3). The test uses a NuWro overlay with the full detector simulation for FHC and RHC. The overlay is first validated against standalone NuWro on the

same flux: its signal rate per POT must agree with the standalone prediction. Pseudo-data thrown from it are then unfolded with the nominal response for every primary and secondary result and compared with the A_C-smeared NuWro truth. The test passes if every bin holding at least 10% of the largest truth lies within 1σ of its total uncertainty and every integral within its uncertainty; the signal region is not opened unless it passes. The one existing NuWro overlay, for Run-1 FHC, cannot serve (Sec. 13.3 of the analysis note).

Proton-tagged results. The proton-tagged secondary results require the inclusive prerequisites U1–U3 and U5–U7, U4 for $\theta_p \times \delta p_T$, and the proton-tagged items listed for each in Table 2. The inclusive control regions do not validate the dominant proton-candidate misidentification of this sample (Sec. 8 of the proton-tagged note). Until an inverted-proton-PID region is defined and validated (P1), no proton-tagged result is unblinded, and θ_p, $\theta_{\pi p}$ and $\theta_p \times \delta p_T$, whose migrations are dominated by the proton-candidate assignment, carry the additional term of P2. The reconstruction of W_{had} and p_n needs a truth-level validation for pion production or a redefinition of the observables as the proxies they are (P4). The model sensitivity of the low δp_T and p_n bins needs an explicit model term in the covariance, the removal of those bins, or an independent-generator test that shows no such shift (P3).

Reporting decision (R1). Every one-dimensional result is quoted per beam mode and for the combination. The responses of the two two-dimensional results can be inverted only in the combined configuration, and whether a result available only there is acceptable is a collaboration decision.

Table 4: Prerequisites of the analysis, with their criteria, owners and evidence, and the stage before which each is required. U: all results; P: proton-tagged results; V: results proposed since the last validation round; R: reporting decisions.

ID	Prerequisite	Criterion	Owner; evidence	Required before
U1	Absolute flux normalisation	Written confirmation that the integrated $\nu_\mu + \bar{\nu}_\mu$ fluxes above 60 MeV used for the normalisation (FHC 6.81159, RHC 6.44646, combined 6.59691 $\times$ 10^{-10} cm^{-2} POT^{-1}) are the PPFX-corrected central values that correspond to the per-event PPFX weights	NuMI flux group (flux contact); written statement	Unblinding
U2	Control regions	The four inclusive control regions compared with beam-on data under the frozen procedure, and any discrepancy treated as that procedure prescribes	Analysers and conveners; control-region comparison	Unblinding
U3	Independent-generator closure	A NuWro overlay with the full detector simulation, validated against standalone NuWro on the same flux, unfolded with the nominal response: every bin holding at least 10% of the largest truth within 1σ of its total uncertainty, and every integral within its uncertainty	Analysers, with the production group for the overlay; closure tables for every primary and proposed result	Unblinding
U4	MCS momentum	Comparison of the MCS and range momenta of contained muons and of the MCS momentum of exiting muons in data and simulation: each difference calibrated or within the MCS term of the covariance	Analysers; control-region MCS study	Unblinding of results that use the muon momentum
U5	Beam-on exposure	POT and trigger counts of every beam-on file summed over the runs the file contains, and checked against the beam database to better than 1%	Analysers, with the beam-data contact; per-run exposure table	Unblinding
U6	Reproduction	The result set rebuilt from empty caches reproduces every released bin to 10^{-9}	Analysers; rebuild log	Unblinding

ID	Prerequisite	Criterion	Owner; evidence	Required before
U7	Terms outside the covariance	The beamline-geometry flux term (1.6–3.3%), the beam-off gate-ratio term and the Run-2 beam-off stand-in term included in the released covariance	Analysers; released covariance	Publication
P1	Proton misassignment control region	An inverted-proton-PID control region at exactly one pion candidate, validated with pseudo-data and compared with beam-on data under the frozen procedure; the rate of wrong proton candidates in the simulation consistent with the data within its uncertainty, or corrected under that procedure	Analysers; region definition and comparison	Unblinding of every proton-tagged result
P2	Wrong-candidate term for the proton angles	The fraction of wrong proton candidates varied by its uncertainty from P1, and the resulting change of every bin of θ_p, $\theta_{\pi p}$ and $\theta_p \times \delta p_T$ added to their covariance	Analysers; covariance term	Unblinding of θ_p, $\theta_{\pi p}$, $\theta_p \times \delta p_T$
P3	Model sensitivity of the low-imbalance bins	With an explicit model term in the covariance, the alternative-model bias of the low δp_T and p_n bins below 1σ; or those bins removed; or U3 showing no such shift	Analysers; covariance term or closure	Unblinding of $\theta_p \times \delta p_T$; promotion of δp_T, p_n
P4	Reconstruction validity of W_{had} and p_n	For pion production, a truth-level study showing that each reconstructed proxy follows the quantity it is named for (hadronic invariant mass, initial-nucleon momentum) with a bias smaller than the bin width in every bin; otherwise the observable redefined as the proxy, with the same prescription at truth level	Analysers; truth-level study	Promotion of W_{had}, p_n
V1	Validation of the newly proposed results	Ensembles (pull widths consistent with one within the ensemble precision, nominal coverage) and the alternative-model test (per-bin bias below 1σ, or flagged and treated) for θ_p, $\theta_{\pi p}$ and the two two-dimensional results	Analysers; ensemble and alternative-model tables of the two notes	Complete (2026-09-28, repeated for version 1.7, reunfolded for version 1.8, $\theta_p \times \delta p_T$ repeated for version 1.9): pull widths 0.92–1.04, largest bias 0.91σ
R1	Results in the combined configuration only	A collaboration decision on quoting the two two-dimensional results only in the combined configuration, since their FHC and RHC responses cannot be inverted	Conveners; decision recorded	Approval of the two-dimensional results

5 Control regions: request and sign-off

The decision requested of the collaboration is permission to inspect beam-on data in four reconstruction-level control regions of the inclusive CC $1\mu 1\pi Xp$ selection while the signal region stays blind. Sec. 10.1 of the analysis note gives the region definitions and Sec. 19 of the technical supplement their roles.

Analysis state. Repository tag `cr-freeze-2026-09-06`; pseudo-data: one Poisson throw per run of the central-value prediction with the unified weight rule (Sec. 5 of the analysis note); binning: the released edges of Appendix A of the analysis note (five-bin forward-split $\cos\theta_\pi$, the p_π binning then in use); normalisation: per-run exposure, per-run beam-off gate ratios, per-configuration flux; detector variations: native Run-4 FHC and RHC samples; data release: `report/data_release/`, checksum manifest `MANIFEST.sha256` (`b8ebc69e3bc91434`).

Request. We request approval to inspect beam-on data in four control regions: CC0π, multi-π, π^0 and large muon–pion opening angle (called "cosmic" below for brevity: it is cosmic-enriched, 34–41% beam-off, not cosmic-pure). Throughout, "beam-off", "EXT" and "cosmic data" name the same data-driven component, the beam-off trigger sample scaled by gate counts; "signal contamination" means the inclusive signal unless the proton-tagged selection is named. Each region is anchored by the inversion of one nominal signal requirement; requirements that become undefined once that inversion is made, or that would sculpt the targeted control sample, are dropped as listed in Table 25 of the technical supplement. Each region is disjoint from the signal region at reconstruction level (demonstrated event by event, Sec. 20.1 of the technical supplement); their expected signal contaminations are 8–24%. The signal region remains blind. Before the regions are opened, the selections, the run-dependent normalisation, the beam-off scaling, the displayed observables and binning, the covariance construction, the goodness-of-fit tests and the permitted responses to a discrepancy are frozen as written in Sec. 20.5 of the technical supplement. Control-region data will be compared with the complete prediction (neutrino MC + beam-off + dirt) separately in FHC and RHC, with the combined configuration as a cross-check. Any discrepancy beyond the pre-declared criteria requires a documented treatment and renewed approval before the signal region is opened. The proton-tagged control regions are not part of this request (Sec. 20.6 of the technical supplement).

Points the collaboration is asked to ratify. (i) The π^0 definition without the final multiplicity cut (Table 25 of the technical supplement); the comparison with the version retaining it is in Sec. 20.4 of the technical supplement and will not be used to modify the region after beam-on inspection. (ii) The π^0 target class CC+NC π^0, adopted by the analysers (the NCπ^0 events the shower veto is also meant to remove had been counted under "NC and ν_eCC"; ν_μCC π^0 alone is 46% of the region), together with the region's diagnostic-only status for this opening. Purity is defined throughout as the target class's share of the neutrino-MC prediction in the region, beam-off and dirt being separately constrained components; both the class definition and the denominator were settled after the purities were known, which is one reason the region is diagnostic-only. (iii) The roles of the other three regions as stated above, and the frozen procedure of Sec. 20.5 of the technical supplement. The collaboration is also asked to ratify the frozen π^0 definition with the comparison of Sec. 20.4 of the technical supplement in view.

Approval within the procedure. The analysers propose the procedure of Sec. 20.5 of the technical supplement for ratification with the approval. It is frozen, and any departure from it is itself a change requiring approval. The exclusion of a run period (item 4(i)) requires a new frozen release and explicit approval. The π^0 region is diagnostic only for this opening: approval to inspect it does not authorise a fitted correction, and a fitted π^0 correction may be proposed only after the pre-fit result is documented, with the fit model, the leverage of the fixed components, the transfer-factor uncertainty and its floor reviewed by the collaboration before the correction is propagated to the signal region; a CC0π correction is pre-authorised under the rule of item 4(ii). If the post-treatment test still fails (item 4(iv)), the background model is revised and renewed collaboration approval is required before the signal region is opened. Conditioning the prediction on the control-region totals, which on an Asimov dataset reduces the uncertainty of the one-bin total from 35.8% to 11.3% (Sec. 10 of the analysis note), would change the role of the control regions and requires a collaboration decision.

Sign-off. The analysis conveners, the flux and detector-systematics contacts, and the collaboration review committee, on the basis of a short report containing the comparisons of item 1 and the response chosen from item 4 of the procedure.

Proton-tagged regions. The proton-tagged control regions are not part of this request: they apply the proton tag rather than inverting it and do not constrain the proton mis-tag background (Sec. 20.6 of the technical supplement), so the proton-tagged control-region programme is not ready for equivalent approval.

Approval checklist.

Item	Status / evidence	Sign-off
Required before the control regions are opened		
Control-region definitions frozen	Table 25 of the technical supplement	
Signal-region overlap zero, event by event	Sec. 20.1 of the technical supplement	
Plot list and binning frozen	item 1 of Sec. 20.5 of the technical supplement	
Pre-fit covariance, global statistic and thresholds frozen	items 2–3 of Sec. 20.5 of the technical supplement; null calibration Fig. 8 of the technical supplement	
Response to a discrepancy frozen	item 4 of Sec. 20.5 of the technical supplement	
π^0 definition and target class ratified; diagnostic-only status recorded	points (i)–(ii), this section	conveners
Flux and detector-systematics contacts have reviewed their parts	pending	contacts
Repository tag and release checksum recorded	analysis state, this section	
Required later, before the signal region is opened		
Control-region results reviewed; any discrepancy treated under Sec. 20.5 of the technical supplement	pending	conveners
Model-distortion pseudo-data closure (reweighted GENIE, FHC)	done, Sec. 13 of the analysis note	
Model-distortion pseudo-data closure, proton-tagged family (FHC)	done, Sec. 5.3 of the proton-tagged note; low-imbalance $\delta p_T/p_n$ bins biased 1.02–1.19σ under GENIE u545, the lower W_{had} region +1.04σ under the angular variation	
Statistical-coverage ensembles, proton-tagged family (W_{had}, δp_T, p_n)	done, Sec. 5.2 of the proton-tagged note; pull widths 0.89–1.04, integral offsets within 1.4%	
Ensembles and alternative-model closure of the results proposed since then, θ_p, $\theta_{\pi p}$, $\theta_p \times \delta p_T$ and $\cos\theta_\pi \times \cos\theta_\mu$ (prerequisite V1)	done, Sec. 5.2 of the proton-tagged note, Sec. 5.3 of the proton-tagged note, Sec. 14.7 of the analysis note; pull widths 0.92–1.04, largest bias 0.91σ	
Independent-generator pseudo-data closure (prerequisite U3): a NuWro overlay with the full detector simulation, validated against standalone NuWro on the same flux, unfolded with the nominal response	pending: the existing NuWro overlay failed validation against the standalone prediction and is not used, Sec. 15 of the analysis note	
Flux constants confirmed as PPFX-corrected central values	pending, Sec. 4.4 of the analysis note	flux contact
MCS calibration of exiting muons to the control regions, applied to every simulated sample, with its uncertainty released as `cov_MCScal.txt` (replacing the ±5% momentum-scale term)	done, Sec. 12.3 of the analysis note; for p_μ at most 0.28σ (FHC), 0.21σ (RHC) and 0.43σ (combined, highest bin); exactly zero for the observables that do not use the muon momentum; computed outside the framework and part of the official covariance (`cov_total_plusMCS.txt`) and of every summary table; prerequisite U4 complete; proton-tagged observables at most 0.35σ (combined p_μ;	